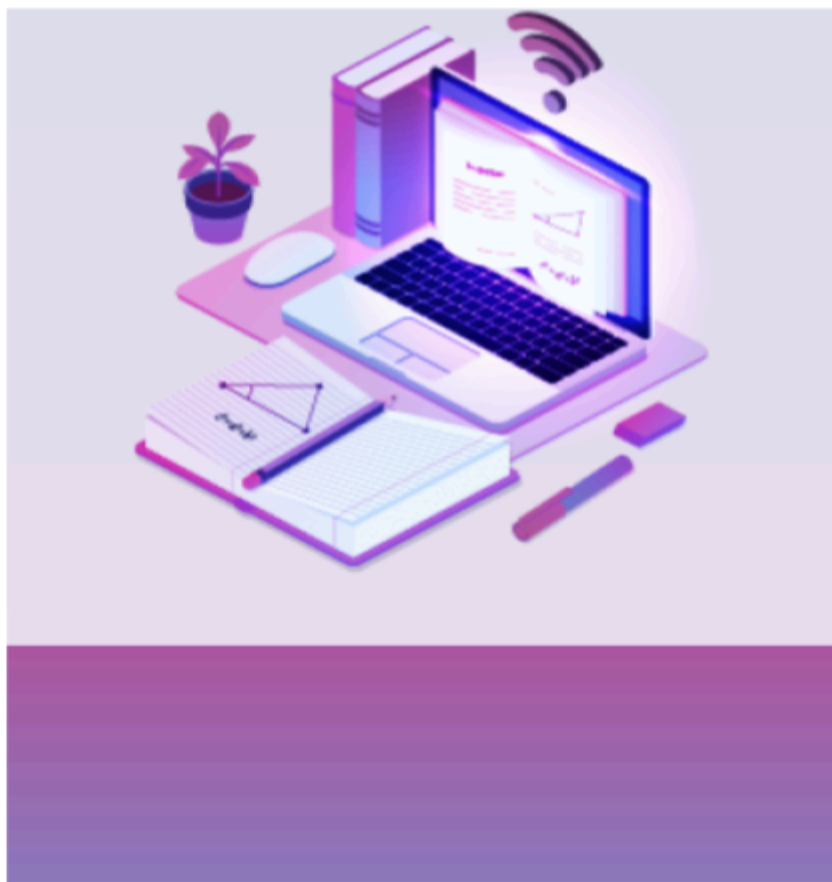


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CAT Logarithms, Surds & Indices Formulas PDF

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CAT Logarithms, Surds & Indices Formulas

- “Logarithms, Surds and Indices” is one of the easiest topics in the quantitative section of the CAT exam.
- Although the number of formulas is high, the basic concepts are very simple to understand and apply.
- There are no shortcuts to remember and the scope of the questions that can be asked is very limited.
- The accuracy of answering questions from this section is very high and good students tend to score very well here.
- If $X, Y > 0$ and m, n are rational numbers then

$$\rightarrow X^m \times X^n = X^{m+n}$$

$$\rightarrow X^0 = 1$$

$$\rightarrow \frac{X^m}{X^n} = X^{m-n}$$

$$\rightarrow (X^m)^n = X^{mn}$$

$$\rightarrow X^m \times Y^m = (X \times Y)^m$$

$$\rightarrow \frac{X^m}{Y^m} = \left(\frac{X}{Y}\right)^m$$

$$\rightarrow X^{-m} = \frac{1}{X^m}$$

- If X and Y are positive real numbers and a,b are rational numbers

$$\rightarrow \left(\frac{X}{Y}\right)^{-a} = \left(\frac{Y}{X}\right)^a$$

$$\rightarrow X^{1/a} = \sqrt[a]{X}$$

$$\rightarrow X^{a/b} = \sqrt[b]{X^a}$$

$$\rightarrow \sqrt[a]{X} \times \sqrt[a]{Y} = \sqrt[a]{XY}$$



$$\rightarrow \frac{\sqrt[a]{X}}{\sqrt[a]{Y}} = \sqrt[a]{\frac{X}{Y}}$$

$$\rightarrow \frac{1}{\sqrt{N+1}-\sqrt{N}} = \sqrt{N+1} + \sqrt{N}$$

- Surds is an irrational number involving a root.

$$\text{Ex: } \sqrt{5}, \sqrt[3]{7}, \sqrt[5]{2}$$

- Like surds are two surds having the same number under radical sign.
- Like surds can be added or subtracted.

$$6\sqrt{2} + 3\sqrt{2} = 9\sqrt{2}$$

- If $a + \sqrt{b} = c + \sqrt{d}$, then $a = c$ and $b = d$.
- The conjugate of $a + \sqrt{b}$ is $a - \sqrt{b}$

$$\bullet \sqrt{a\sqrt{a\sqrt{a\ldots\infty}}} = a$$

$$\bullet \sqrt{a\sqrt{a\sqrt{a\ldots x \text{ times}}}} = a^{1-\left[\frac{1}{2^x}\right]}$$

- To find $\sqrt{\sqrt{x} + \sqrt{y}}$, $\sqrt{x} + \sqrt{y}$ should be written in the form of $m + n + 2\sqrt{mn}$ where $x = m + n$ and $4mn = y$ and $\sqrt{\sqrt{x} + \sqrt{y}} = \pm (\sqrt{m} + \sqrt{n})$
- If $N = a^x$ then, x is defined as the logarithm of N to base a or $x = \log_a N$ a logarithm of a negative number or zero is not defined
- $\log_a 1 = 0$
- $\log_a xy = \log_a x + \log_a y$
- $\log_a b^c = c \log_a b$
- $\log_a a = 1$
- $X^{\log_b y} = Y^{\log_b x}$



- $\log_a \sqrt[n]{b} = \frac{\log_a b}{n}$
 - $\log_a x = \frac{1}{\log_x a}$
 - $b^{\log_b x} = x$
 - $\log_a b = \frac{\log_c b}{\log_c a}$
 - $\log_a b * \log_b a = 1$
 - $\log_a \left(\frac{X}{Y} \right) = \log_a X - \log_a Y$
 - If $0 < a < 1$, then $\log_a x < \log_a y$ (if $x > y$)
 - If $a > 1$ then $\log_a x > \log_a y$ (if $x > y$)
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